

syms K_p T_n T_1 T_2 v s

forward_path = K_p * (1 + 1 / (s * T_n)) * (v / (1 + s * T_1)) * (1 / (1 + s * T_2))

forward_path =

$$\frac{K_p v \left(\frac{1}{T_n s} + 1 \right)}{(T_1 s + 1) (T_2 s + 1)}$$

feedback_path = 1

feedback_path = 1

TF = forward_path / (1 + forward_path * feedback_path)

TF =

$$\frac{K_p v \left(\frac{1}{T_n s} + 1 \right)}{(T_1 s + 1) (T_2 s + 1) \left(\frac{K_p v \left(\frac{1}{T_n s} + 1 \right)}{(T_1 s + 1) (T_2 s + 1)} + 1 \right)}$$

[num, den] = numden(TF);

num_poly = collect(num, s);

den_poly = collect(den, s);

TF_polynomial = num_poly / den_poly

TF_polynomial =

$$\frac{(K_p T_n v) s + K_p v}{(T_1 T_2 T_n) s^3 + (T_1 T_n + T_2 T_n) s^2 + (T_n + K_p T_n v) s + K_p v}$$

num_poly

num_poly = (K_p T_n v) s + K_p v

den_poly

den_poly = (T_1 T_2 T_n) s^3 + (T_1 T_n + T_2 T_n) s^2 + (T_n + K_p T_n v) s + K_p v